Category Theory

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Introduction

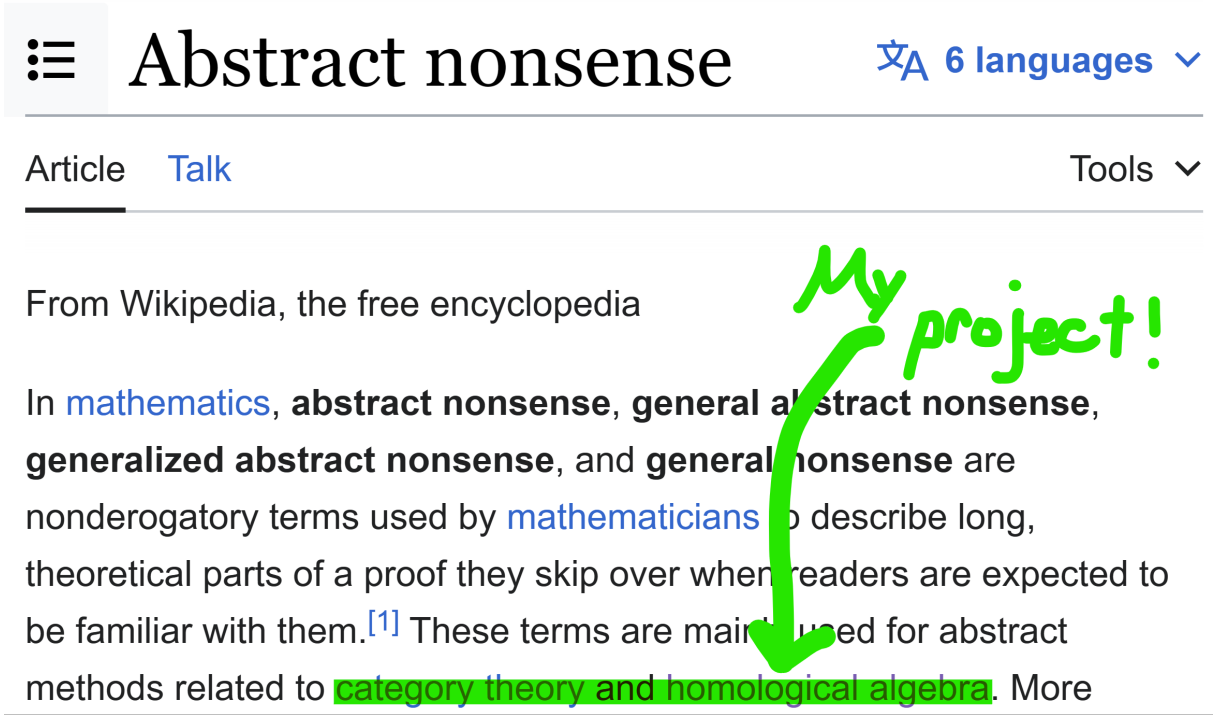


Figure: Wikipedia, *Abstract Nonsense*.

Table of Contents

1. The Basics
   - Definition and Examples
   - Functors
2. The Categorical Perspective
   - Dualizing
   - Universal Properties
3. Looking Ahead

Definition

Motivation (Philosophically): Looking for some definition that captures all the "structures" in all sets...

Definition

A **category** $\mathscr{C}$ contains 4 pieces of information:

- **Objects** $\mathrm{Ob}(\mathscr{C})$.
- Given two objects A, B, a set of **Morphisms** between them $\mathrm{Hom}_{\mathscr{C}}(A, B)$, an element of this set is usually written $f : A \to B$.
- For each object $A \in \mathrm{Ob}(\mathscr{C})$, the **identity morphism** $1_A : A \to A$.
- A **composition rule** to produce $g \circ f : A \to C$ from $f : A \to B, g : B \to C$.

Examples - Usual Ones

Category	Objects	Morphisms
Set	Sets	Functions
Grp	Groups	Group Homomorphisms
Ab	Abelian Groups	Group Homomorphisms
$\mathbf{Vect}_{\mathbb{C}}$	Vector spaces over $\mathbb{C}$	Linear Maps
Top	Topological Spaces	Continuous Maps
Ring	Commutative Rings with unity	Ring Homomorphisms
Fld	Fields	Field Homomorphisms

Table: A table of common categories.

Example, not so usual ones

Here is a category $\mathscr{C}$.

- Objects: $\{0, 1, 2, 3, \cdots\}$
- Morphisms:

$$\mathrm{Hom}_{\mathscr{C}}(x, y) = \begin{cases} \{*\} & \text{if } x \leq y \\ \emptyset & \text{otherwise} \end{cases}$$

called the **poset categories**.

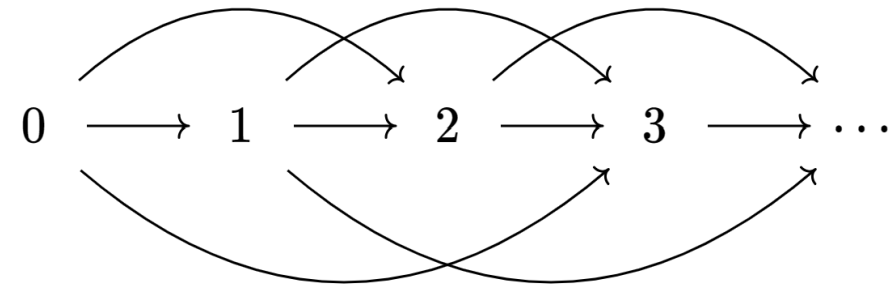


Figure: A Visualization of the the poset category $(\mathbb{N}, \leq)$.

Example, not so usual ones (2)

Given a group G, it is a category itself (not $\mathbf{Grp}$)!

- Objects: A single one (call it $*$).
- Morphisms: elements of G.
- Composition rule: group multiplication.

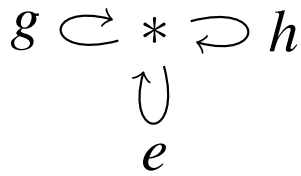


Figure: A visualization of the Category G.

Functors

Definition

A **functor** from $\mathscr{C}$ to $\mathscr{D}$, denoted by $F : \mathscr{C} \to \mathscr{D}$ is a specified way (two functions) that sends

- Objects A in $\mathscr{C}$ to Objects $F(A)$ in $\mathscr{D}$, and
- Morphism $f : A \to B$ to Morphism $F(f) : F(A) \to F(B)$

That satisfies $F(g) \circ F(f) = F(g \circ f)$ and $F(1_A) = 1_{F(A)}$.

Examples of Functors

- A functor from the above category $\mathscr{C}$ to $\mathscr{C}$ is a order-preserving function.
- A functor from G to H (as two categories) is a group homomorphism.
- A functor between G and $\mathbf{Set}$ is a group action.

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Dualizing

In this section, fix a category $\mathscr{C}$.

Definition

The **opposite category** is a category $\mathscr{C}^{op}$ in which we define the objects and identities as the same, but $\mathrm{Hom}_{\mathscr{C}^{op}}(A,B) = \mathrm{Hom}_{\mathscr{C}}(B,A)$ and define composition accordingly.

Examples of Dualizing

Definition

A morphism $f : A \to B$ is an **mono** if $f \circ g_1 = f \circ g_2$ implies $g_1 = g_2$ for any $g_1, g_2 : X \to A$ (for any object X).

Dualizing:

Definition

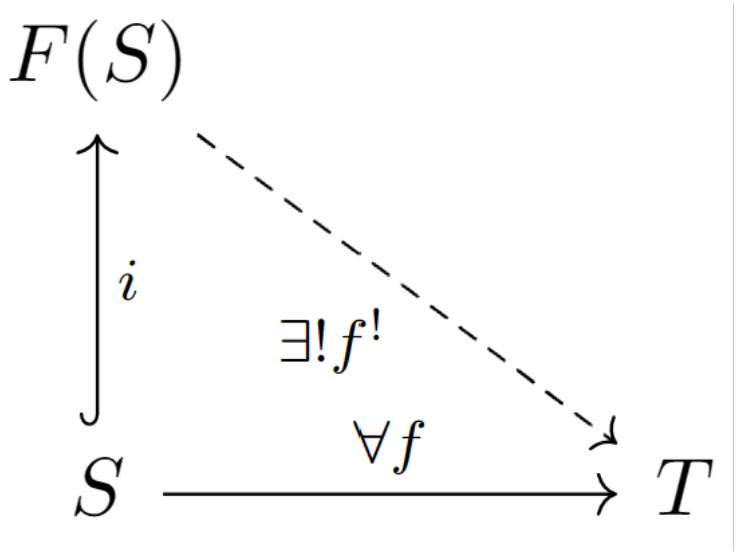
A morphism $f : A \to B$ is an **epi** if $g_1 \circ f = g_2 \circ f$ implies $g_1 = g_2$ for any $g_1, g_2 : B \to X$ (for any object X).

Universal Property (1)

Definition

Given a set S, the **free vector space** $F(S)$ such that it satisfies the following two properties:

- There is an injective function $i : S \to F(S)$, and
- For any vector space T, if there is a function $f : S \to T$, then there exists a unique linear map $f^! : F(S) \to T$ with $f^! \circ i = f$.



Universal Property (2)

- We gave a **basis-free definition** above!
- This is because we only specify what the *morphisms* do, this is a really powerful concept!

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Other Topics

- Special objects and functors: Adjoints functors, Limits...
- Application to Algebra: (Bounded) quiver algebras, indecomposable simple, projective, injective kQ/I-modules
- More structures on Categories: Localization of Categories, Triangulated Categories...
- The "injective generates" condition: finitistic dimensional conjecture...

Thank You!

Any questions?